Simplification Rules

Inference:-

In artificial intellegence we need intelligent computers which can create new logic from old logic or by evidence, so generating the conclusions from evidence and facts is termed as inference.

Inference rules:-

Inference rules are the templates for generating valid arguments. Inference rules are applied to derive proofs in artificial intelligence, and the proof is the sequence of the conclusion that leads to the desired goal.

Terminologies of inference rules:-

→ Impli cation

→ Converse

→ Contrapositive

→ Inverse

Implication:-

It is one of the logical connectives which can be represented as $p \to q$.

It is a boolen expression.

p	q	$p \to q$
T	T	T
T	F	F
F	T	T
F	F	T

Converse:-

The converse of implication, which means the right hand proposition goes to the left hand side. It can be written as $q \to p$ and it can be read as q implies p.

p	q	$q \to p$
T	T	T
T	F	T
F	T	F
F	F	T

Contrapositive:-

The negation of converse is termed as contrapositive, and it can be represented as $\sim q \to \sim p$.

p	q	$\sim q$	$\sim p$	$\sim q \to \sim p$
T	T	F	F	T
T	F	T	F	F
F	T	F	T	T
F	F	T	T	T

Inverse:-

The negation of implication is called inverse. It can be represented as $\sim p \rightarrow \sim q$.

p	q	$\sim p$	$\sim q$	$\sim p \rightarrow \sim q$
T	T	F	F	T
T	F	F	T	T
F	T	T	F	F
F	F	T	T	T

Types of Inference Rules:-

- → Modus ponens
- → Modus Tollens
- → Hypothetical syllogism
- → Disjunctive syllogism
- → Addition
- → Simplification
- → Resolution

Simplification Rules:-

The simplification rule state that if $p \wedge q$ is true the $q \vee p$ will also true.

Example:-

→ It is raining or I will make tea.

→ It is not raining, or I will read a book.

→ Therefore I will make tea, or I will read a book.

Resolution:-

$$((p \vee q) \wedge (\sim p \vee r)) \Rightarrow (q \vee r)$$

p	q	$p \wedge q$
T	T	T
T	F	T
F	T	T
F	F	F

Rewrite Rules:-

To write (something) agin especially in a different way in order to improve it or to include new information.

Example:-

A sentence (S) can be rewritten as a noun phrase (NP) plus a verb phrase (VP) : $S \rightarrow NP + VP$ and the verb phrase can be written as a verb (V) plus a noun phrase (NP): $VP \rightarrow V + NP$. Also called a phrase-structure rule.

Meta Rules:-

Meta rules define knowledge about how the system will work.

Example:-

Meta rules might define that knowledge from Expert A is to be trusted more then knowledge from Expert B.

Meta rules are treated by the system like normal rules, but are given higher priority.